

Signal biclustering across time points reveals persistent center–flank polarization

Biclustering of signal counts across genomic regions and the three measurement channels at 0 hr, 6 hr, and 24 hr revealed a persistent center–flank polarization. At all time points, the dominant biclustered blocks comprise region modules with elevated signal in the central channel and reduced signal in both flanking channels, indicating a stable coupling between center enrichment and flank depletion (Figures 1–3). This canonical pattern appears as multiple contiguous modules at 0 hr (Figure 1), as distinct row blocks with strong central intensities and subdued flanks at 6 hr (Figure 2), and as numerous blocks with the same contrast at 24 hr (Figure 3). The recurrence of these blocks across independently biclustered matrices supports stable modules whose center–flank polarity is maintained over the 24-hour interval.

Superimposed on this dominant architecture are time-specific deviations. At 0 hr, top-ranked modules display weaker or mixed contrasts, including a small block with left-flank elevation and a low right flank, indicating localized baseline asymmetry (Figure 1). At 6 hr, several upper blocks are uniformly low across channels, occasionally with central reduction, consistent with a transient global suppression module not prominent at 0 hr or 24 hr (Figure 2). By 24 hr, a smaller set of modules near the top reverses the canonical polarity—reduced central signal with elevated right flank—and additional blocks show sustained left–right asymmetry, with the right flank frequently higher (Figure 3).

Together, the heatmaps show that most region modules retain central-channel enrichment with concomitant flank depletion at 0 hr, 6 hr, and 24 hr, whereas a minority exhibits time-

restricted behaviors, including uniform suppression at 6 hr and flank-asymmetric or polarity-reversed patterns at 0 hr and 24 hr.

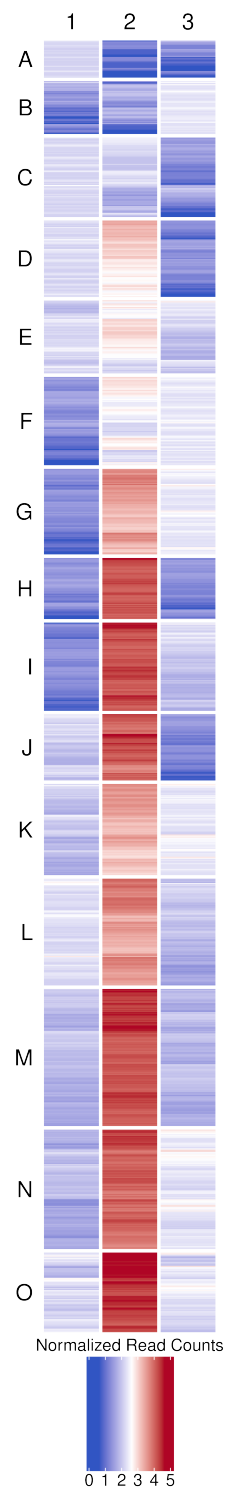


Figure 1. Biclustering reveals central-channel enrichment at 0hr. Rows are genomic regions and columns are paired channels at 0hr. Most modules show higher signals in the central channel with lower signals in the left and right channels, while a few modules display left-channel enrichment or mixed profiles.

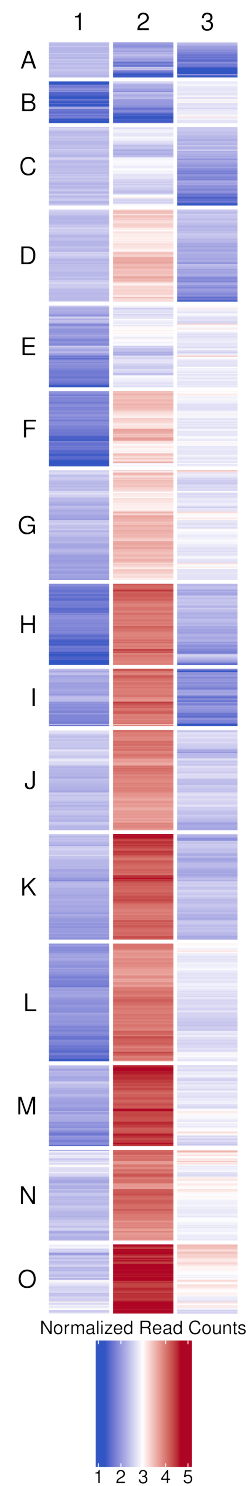


Figure 2. Biclustered 6hr regions show central-channel enrichment across groups. *Heatmap of biclustered signal counts for the 6hr sample with genomic regions as rows and three paired channels as columns. Most biclusters exhibit elevated signal in the center channel with reduced signal in the left and right channels, while a few groups show uniformly low signal.*

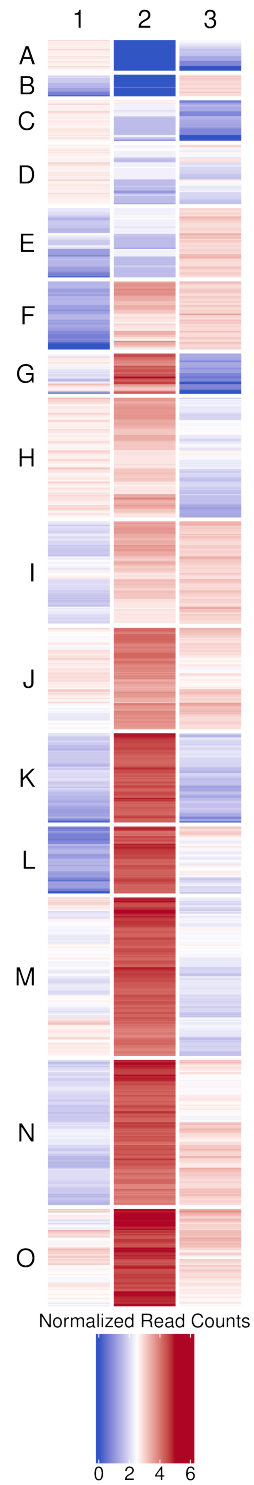


Figure 3. Biclustering reveals dominant center-versus-flanks contrasts at 24hr. *Heatmap of biclustered genomic regions across three measurement channels at 24hr. Most biclusters show higher signal in the middle column with lower signals in both outer columns, whereas a minority display the inverse or an asymmetry favoring the right column.*

Time-resolved TF motif enrichment shifts from selective to broad regulatory engagement

Consistent with the chromatin signal architecture, transcription factor binding site (TFBS) enrichment shifts from baseline cluster-selective patterns to broadly distributed regulatory engagement over time. At 0 hr (Figure 4), enrichment is highly polarized across biclusters: mid-to-right modules (e.g., H–J) show broad gains, whereas leftmost modules (A–D) harbor multiple depletions. GABPA is selectively and strongly enriched in cluster A; KAT2A, E2F4, SIN3A, and RCOR1 are depleted in A–C; USF2, MAX, and TCF12 peak in cluster I; and CTCF remains relatively uniform across clusters. These features indicate a baseline landscape dominated by discrete ETS- and E-box–centered modules superimposed on largely invariant architectural insulation.

By 6 hr (Figure 5), enrichment redistributes toward a broadly positive regime. PAX5 shows the most consistent, high-level signal across biclusters; AP-1 components (JUN, JUND), EP300, and GATA factors are elevated across many clusters with only limited early-cluster depletions. Pol II (POLR2A) and its Ser2-phosphorylated form (Pol II Ser2P), MXI1, NANOG, and ZMIZ1 remain weak in A–C, and cohesin-associated RAD21 is intermediate without sharp peaks. E-box factors (USF1/USF2, MAX) and ETS1/GABPA tend to rise in mid-to-late clusters (approximately H–L). These 6 hr patterns are consistent with rapid engagement of lineage-defining and stimulus-responsive regulators together with a coactivator axis (EP300) that would facilitate enhancer activation (Zhang et al., 2026), while Pol II–associated signals remain comparatively restrained in early clusters.

At 24 hr (Figure 6), enrichment becomes both stronger and more uniform. JUND shows the most pervasive, prominent signal across clusters, with widespread enrichment also for PAX5, EP300, AP-1 partner JUN, cohesin subunits (RAD21, SMC3), MYC, CTCF, TBP, and Pol II (including Pol II Ser2P). Residual localized depletions persist for USF2, E2F4, and ZMIZ1 in early clusters, and pluripotency factors (NANOG, POU5F1) remain weak. The late-time activation of cohesin-associated features is consistent with consolidation of higher-order genome folding and enhancer–promoter communication (Costantino et al., 2026), while concomitant enrichment of TBP and Pol II indicates broad engagement of the basal and productive transcriptional machinery. MYC co-enrichment further supports transcriptional amplification at this stage.

Together, these time-resolved profiles support a model in which: (i) baseline regulation is compartmentalized into selective modules, including a discrete ETS (GABPA) axis against a stable architectural backdrop (uniform CTCF); (ii) by 6 hr, a transition occurs to system-wide enhancer activation driven by PAX5, AP-1 (JUN, JUND), and EP300, accompanied

by rising E-box/ETS activity in later clusters; and (iii) by 24 hr, regulatory control becomes global, marked by sustained AP-1 (JUND), pervasive lineage and coactivator occupancy (PAX5, EP300), recruitment of cohesin (RAD21, SMC3) to stabilize chromatin architecture, and widespread engagement of TBP and Pol II together with MYC, consistent with consolidation of transcriptional programs. The persistence of uniform CTCF across time suggests stable domain boundaries, whereas the progressive rise of cohesin- and Pol II-associated features indicates a shift from localized, factor-specific modules at baseline to

a broadly activated, structurally integrated state by 24 hr. References: Costantino et al., 2026; Zhang et al., 2026.

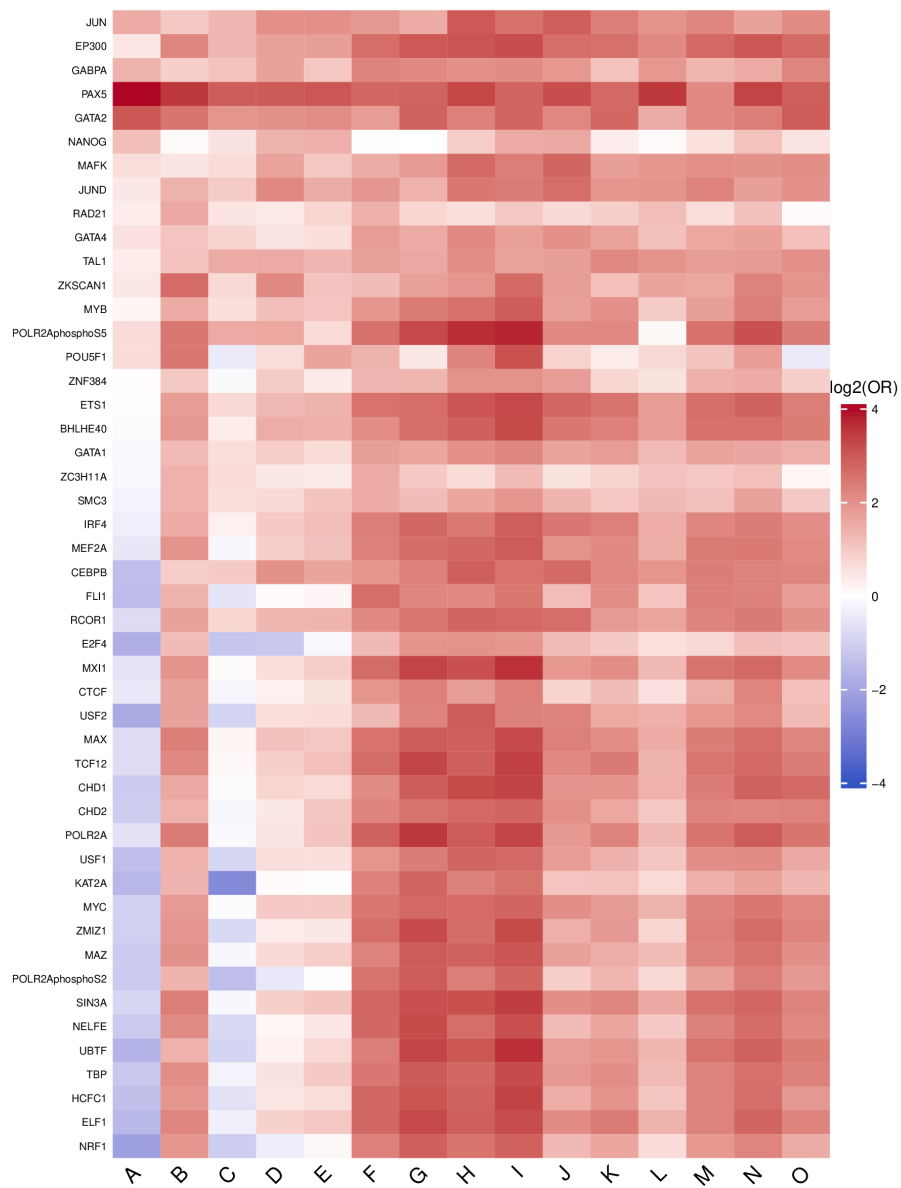


Figure 4. TF motif enrichment across 0hr biclustered genomic regions. Motif overrepresentation ($\log_2$ odds ratio) for known TF binding sites is summarized across region clusters A–O at 0hr. Enrichment concentrates in central-right clusters, with H–J broadly high, while leftmost clusters A–D are weaker or negative, except for a GABPA peak in A.

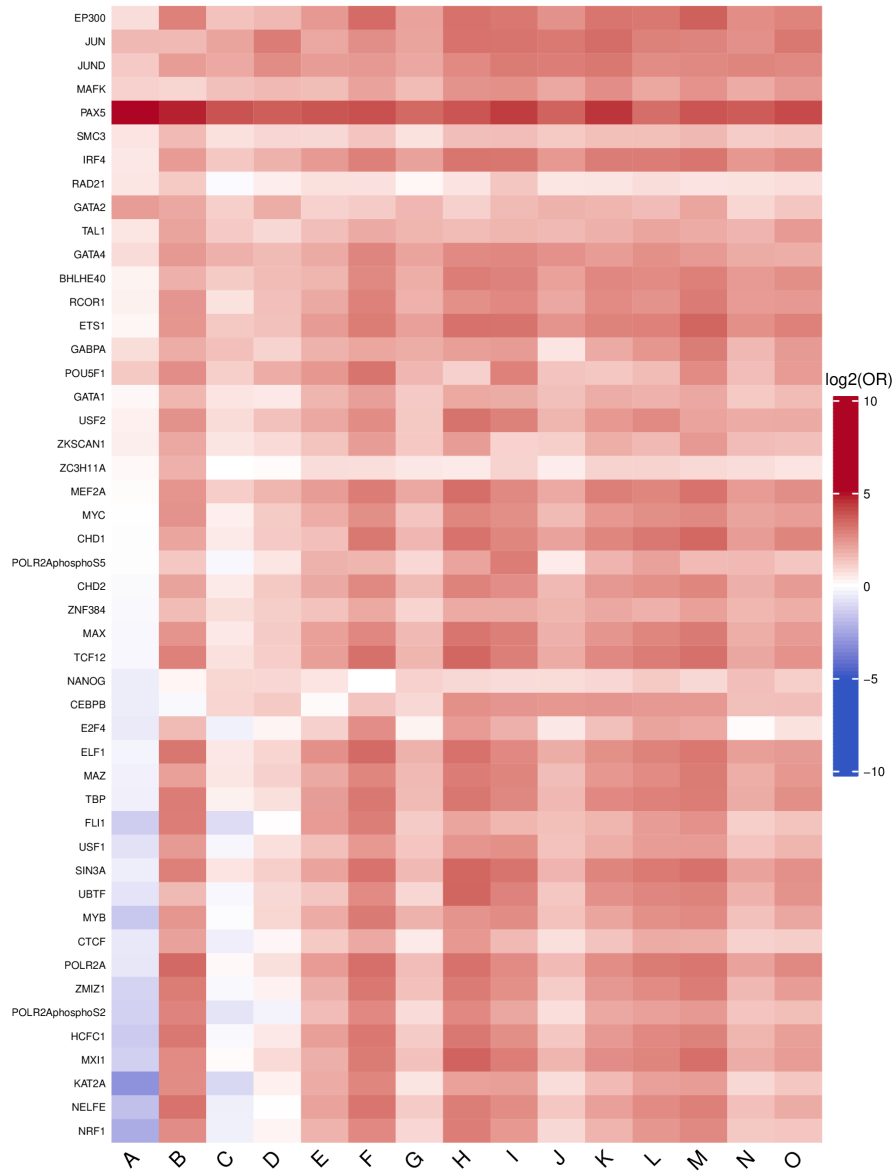


Figure 5. TF motif enrichment across 6hr genomic biclusters. Rows denote TFBS motifs and columns indicate biclusters A–O at 6 hr, with values representing enrichment. Enrichment is widespread, highlighted by consistently high PAX5 across clusters and localized depletion of CTCF, POLR2A, MXI1, and NANOG in early clusters.

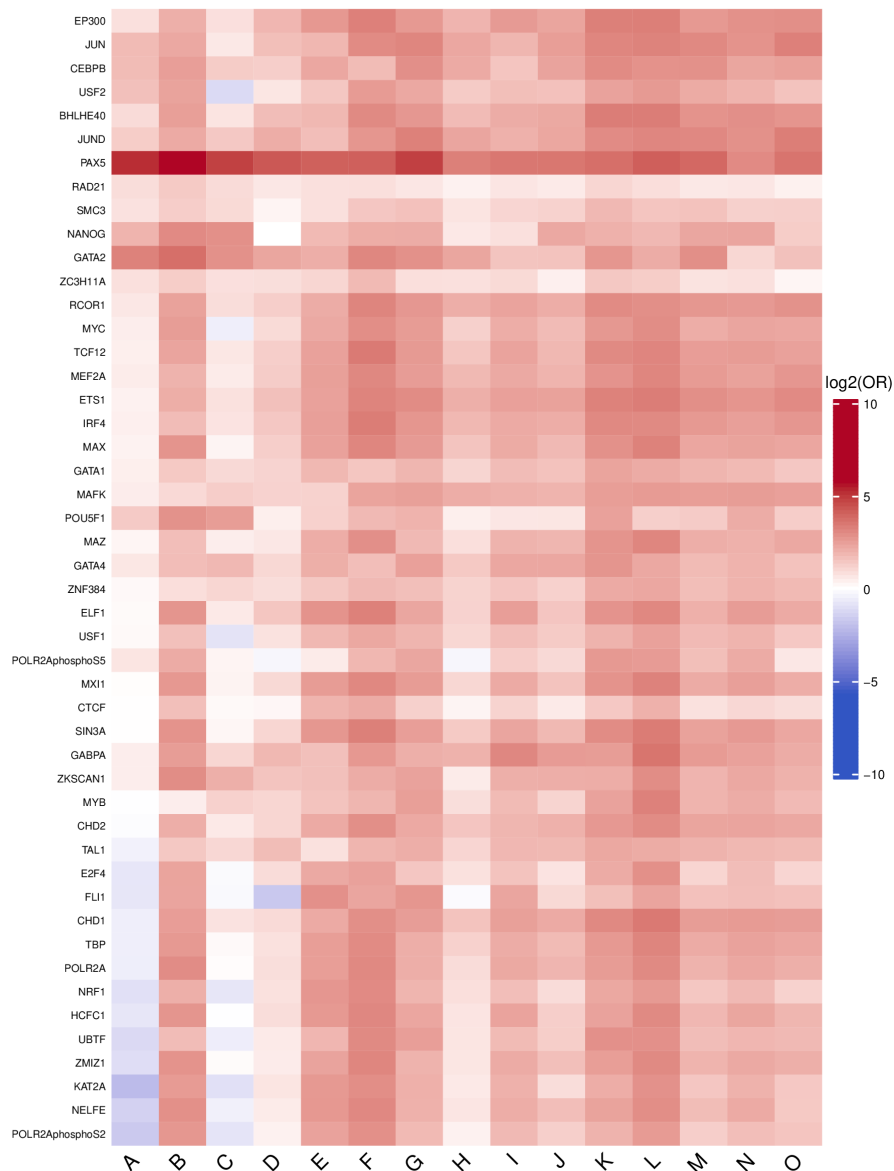


Figure 6. TFBS motif enrichment across 24hr region modules. Motif enrichment is summarized for transcription factors across clusters A–O at 24hr. Signals are broadly positive, with strong and uniform enrichment for JUND and widespread enrichment for factors such as PAX5, EP300, MYC, CTCF, TBP, and POLR2A variants; localized depletions are most apparent for USF2, E2F4, and ZMIZ1 in early clusters.

Regulatory element class composition emphasizes enhancer-dominant clusters

Across time points, cluster-wise cCRE compositions are enhancer dominant, with distal enhancer-like signatures (dELS) comprising the largest fraction and proximal enhancer-like signatures (pELS) typically second most abundant (Figures 7–8). Promoter-like signatures (PLS) remain a minority overall, and chromatin accessibility–associated categories (e.g.,

CA-H3K4me3, CA-CTCF, CA-TF) contribute only modestly in all clusters; an “other” category varies idiosyncratically by cluster.

Time-resolved comparisons highlight cluster-specific redistribution of PLS while preserving the global enhancer skew. At 0 hr, PLS is limited in early clusters (A–F) and becomes more prominent from G onward, with notable contributions in I–N (Figure 7). By 24 hr, PLS increases in selected groups, including B, J, and K, while remaining lower in A and C–D (Figure 8). Thus, a subset of clusters shows a time-dependent rise in promoter-like representation (e.g., B) or sustained/enhanced PLS in mid-to-late clusters (e.g., J–K), yet the predominant enhancer-centric composition ($dELS > pELS \gg PLS$) persists. This pattern is consistent with enhancer-centric regulatory architecture in mammalian cCRE

landscapes, where pELS/dELS elements frequently dominate active repertoires (Konovalov et al., 2025).

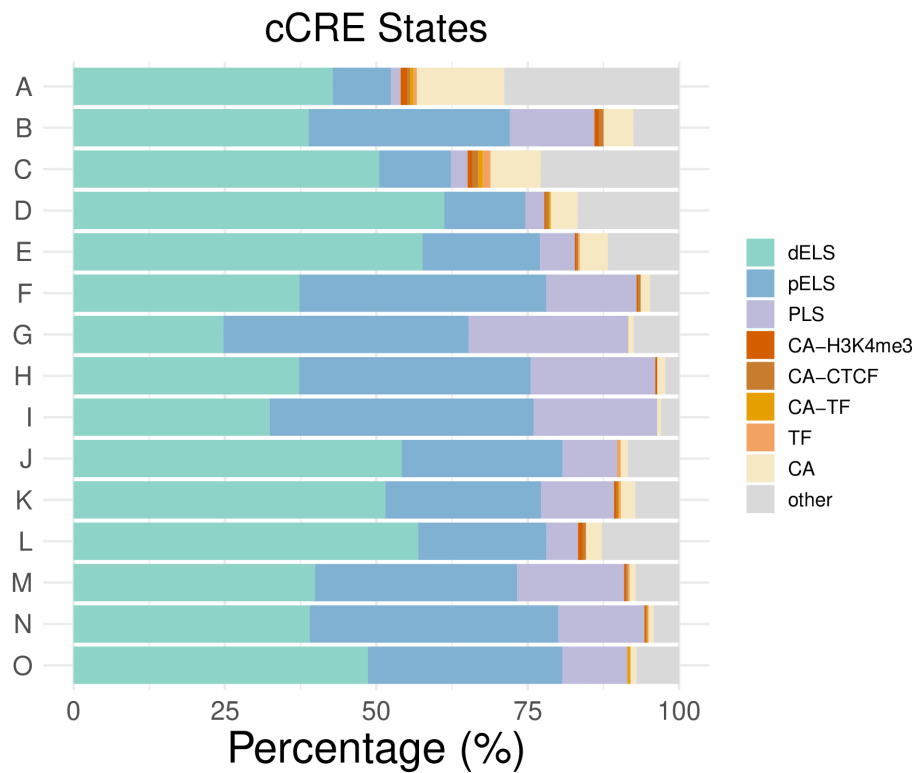


Figure 7. cCRE composition across 0hr clusters highlights enhancer dominance. Clusters A–O are partitioned by ENCODE cCRE classes on mm10 to show the regulatory makeup at baseline. Distal and proximal enhancers constitute the majority in most clusters, while promoter-like states increase in the mid-to-late groups (e.g., G–N); CA/TF-related and other categories form smaller fractions.

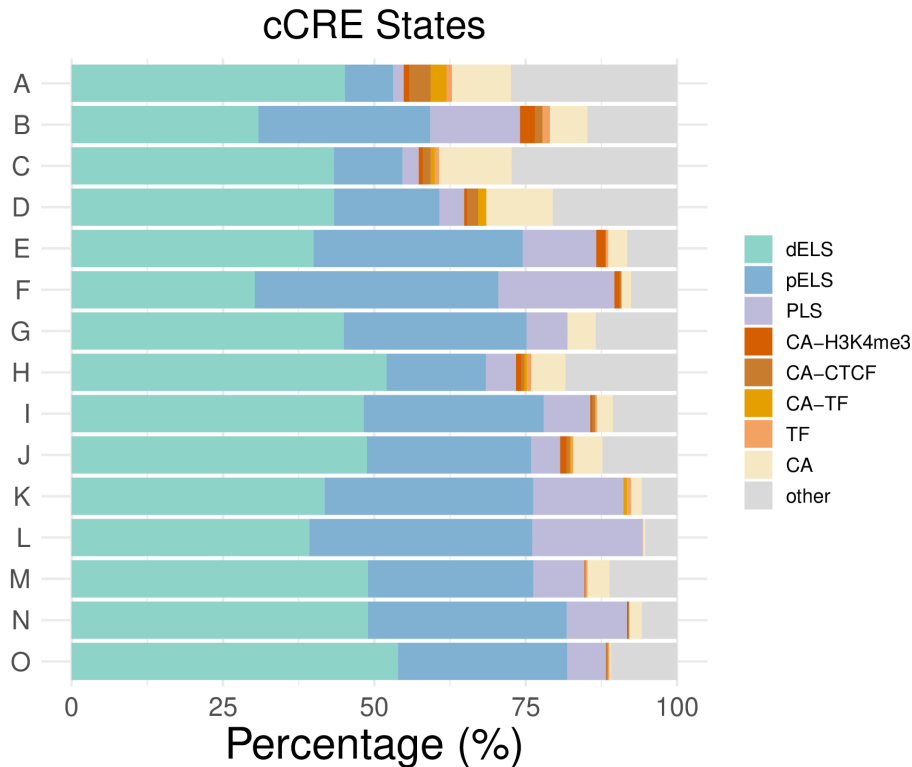


Figure 8. 24hr clusters are enriched for enhancer-like cCRE states. Clusters A–O are assigned to ENCODE cCRE categories, summarizing regulatory element composition. dELS and pELS constitute the majority across clusters, while PLS varies and is comparatively higher in groups such as B, J, and K; CA–H3K4me3/CTCF/TF categories are minor. This distribution outlines the regulatory makeup of the 24hr clusters.

Genic context of clustered regions indicates intron-centric localization

Intersecting clustered regions with mm10 gene models revealed an intron-centric genic context at both time points, with modest, cluster-specific promoter and exon contributions. At 0 hr, intronic annotations predominate across most clusters—most prominently in J, L, and O—while promoter content is consistently lower overall but relatively enriched in B, D, H, and O (Figure 9). Exonic fractions are moderate in several groups (notably E, I, and N), 5' UTRs are minor but more apparent in G and I, and a variable “other” category (including non-coding or unclassified loci) contributes differentially across clusters (e.g., A, C, and K) (Figure 9).

By 24 hr, intronic localization becomes uniformly dominant—the largest category in every cluster—reinforcing a gene body–centered distribution of signals (Figure 10). Promoter content remains the second-largest category but varies markedly by cluster, being relatively low in A, G, and M–N and comparatively higher in C–E, H, and O. Exonic fractions stay minor overall with localized increases (F, I, and N), and both 5' and 3' UTRs remain small components, with 3' UTR contributions detectable as small segments in multiple clusters (e.g., A, F, H, J–M, O) (Figure 10).

Together, these profiles indicate a persistent—and at 24 hr, more consolidated—intron dominance across clusters, with promoter-associated signals present but limited to subsets of clusters and showing stable enrichment in some (e.g., H and O) across time. Exonic fractions display cluster-restricted increases but remain secondary. Overall, the genic

context of the clustered regions is intron-centric at both 0 hr and 24 hr, with modest, cluster-dependent variability in promoter and exon localization.

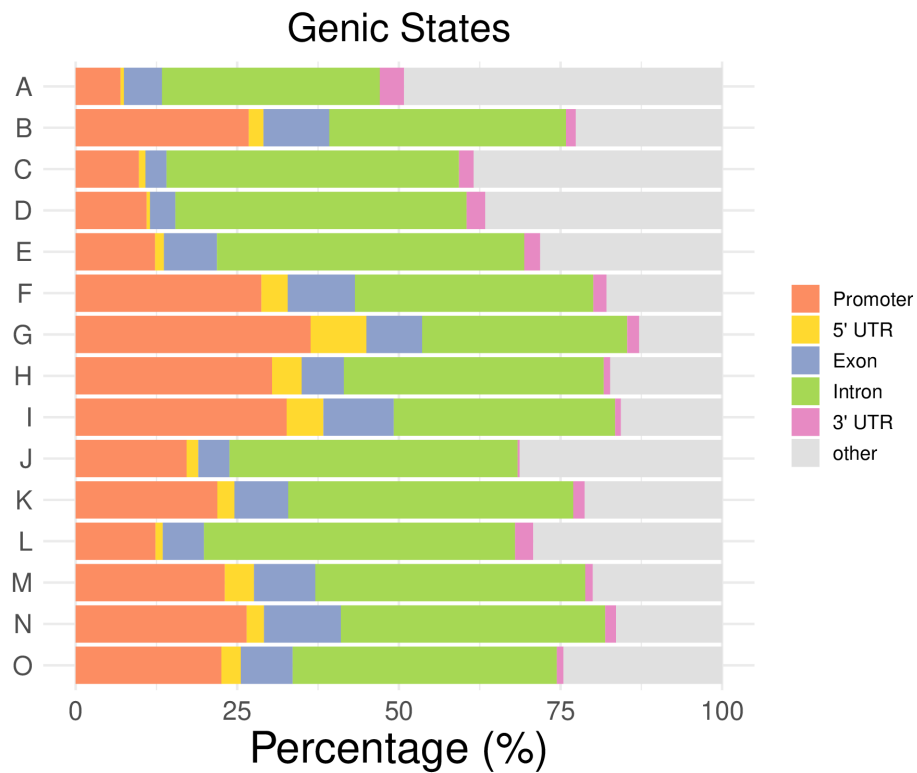


Figure 9. Genic annotation distribution of 0hr region clusters. Region clusters A–O at 0hr are assigned to promoter, 5' UTR, exon, intron, 3' UTR, and other categories by percentage. Most clusters are intron-enriched with smaller promoter contributions; exon and UTR fractions are comparatively minor, and the proportion labeled “other” varies among clusters.

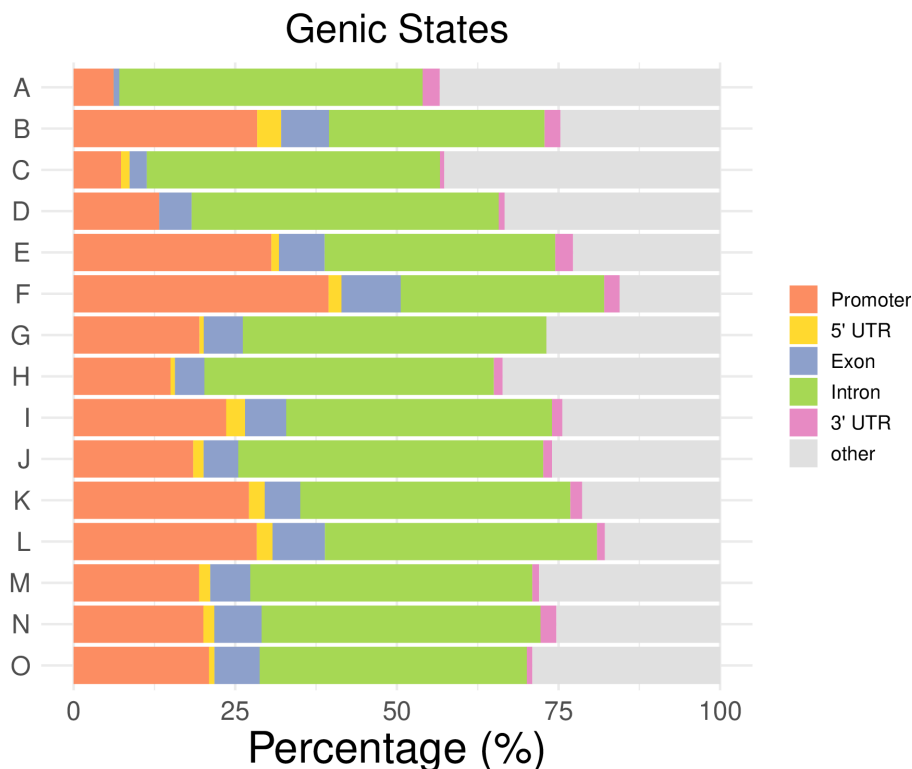


Figure 10. Genic annotation of 24hr clusters shows intron dominance. For clusters A–O at 24 hr, the proportions of promoter, 5' UTR, exon, intron, 3' UTR, and other annotations are compared. Intronic annotations predominate across clusters, while promoter signal is variable and generally second most common. Exon representation is relatively higher in F, I, and N.

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